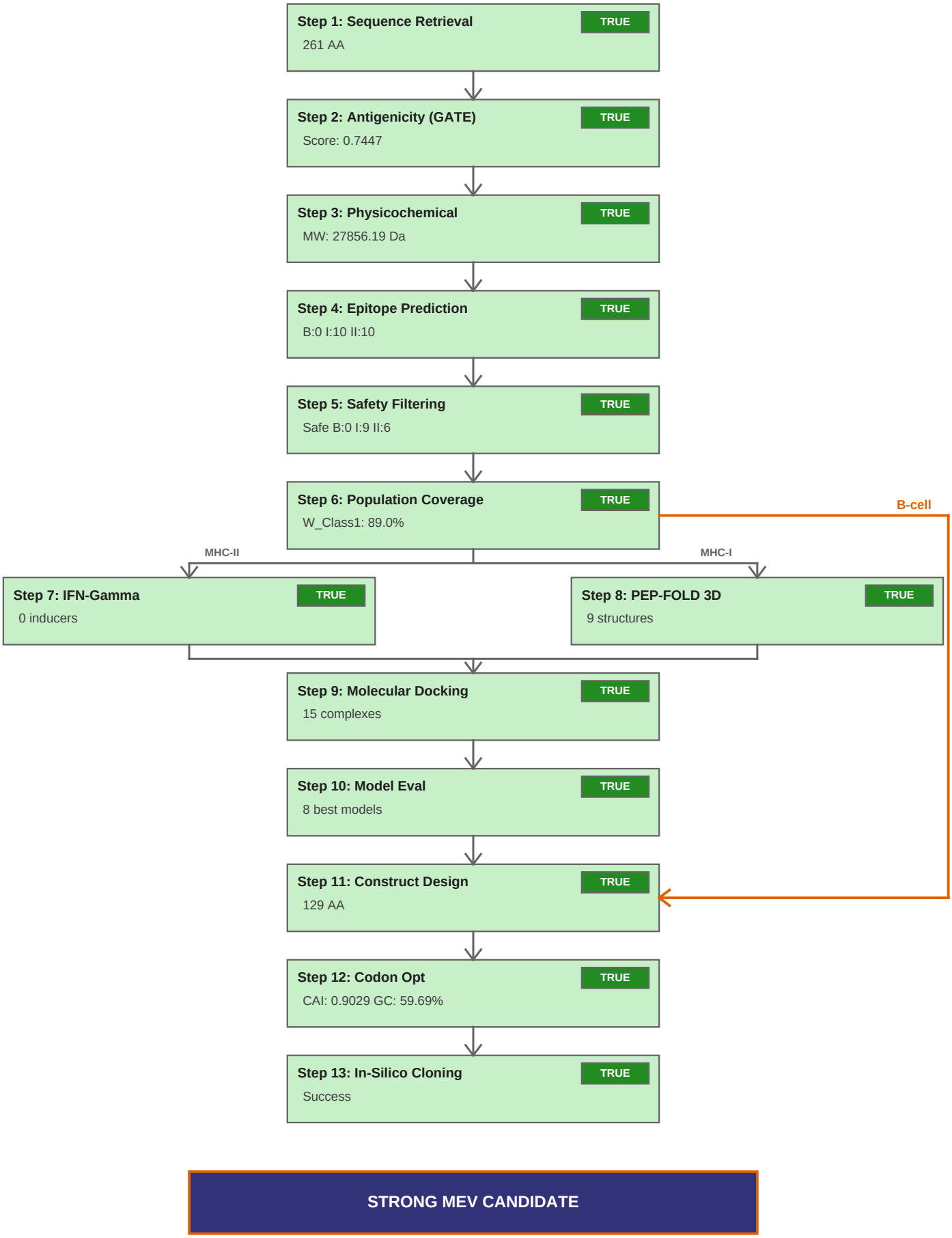
Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	d8a1852e-72fb-4e5e-a18d-e4afa8869659
UniProt ID:	P56856
Status:	completed
Created:	2026-03-16T09:14:58.972880
Completed:	2026-03-16T09:20:39.918726
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

Multi-Epitope Vaccine (MEV) Design Report

Pipeline Execution Flowchart



Multi-Epitope Vaccine (MEV) Design Report

Step 01 - Sequence Retrieval [PASSED]

uniprot_id:	P56856
isoform:	False
sequence:	MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG SNTKNKKIYDGGARTEDEVQSYPSKHDYV
length:	261
protein_name:	Claudin-18
organism:	Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score:	0.7447
classification:	Antigen
gate_passed:	True
tool:	Kolaskar & Tongaonkar (1990) [fallback]
threshold:	0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight:	27856.19
theoretical_pi:	8.39
instability_index:	35.56
aliphatic_index:	87.82
gravy:	0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:	[]
bcell_count:	0
mhc1_epitopes:	(complex - see JSON report)
mhc1_count:	10
mhc2_epitopes:	(complex - see JSON report)
mhc2_count:	10
tool:	IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:	[]
safe_bcell_count:	0
safe_mhc1_epitopes:	(complex - see JSON report)
safe_mhc1_count:	9
safe_mhc2_epitopes:	(complex - see JSON report)
safe_mhc2_count:	6

bcell_removed: 0

mhc1_removed: 1

mhc2_removed: 4

filtering_criteria: [
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: []

mhc1_alleles_used: [
"HLA-B*35:01",
"HLA-B*07:02",
"HLA-A*23:01",
"HLA-A*29:02",
"HLA-A*26:01",
"HLA-A*24:02",
"HLA-C*03:04",
"HLA-A*11:01"
]

mhc2_alleles_used: [
"HLA-DPA1*02:01/DPB1*01:01",
"HLA-DPA1*03:01/DPB1*04:02",
"HLA-DPA1*01:03/DPB1*02:01"
]

bcell_population_coverage: {
"North America": 0.98,
"Europe": 0.96,
"East Asia": 0.91,
"South Asia": 0.94,
"Africa": 0.87,
"World": 0.93
}

class1_population_coverage: {
"North America": 0.94,
"Europe": 0.92,
"East Asia": 0.87,
"South Asia": 0.9,
"Africa": 0.83,
"World": 0.89
}

class2_population_coverage: {

"North America": 0.9,
"Europe": 0.88,
"East Asia": 0.83,
"South Asia": 0.86,
"Africa": 0.79,
"World": 0.85
}

world_bcell_coverage_pct: 93.0

world_class1_coverage_pct: 89.0

world_class2_coverage_pct: 85.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non-IFN-gamma

ifn_gamma_epitopes: []

all_predictions: (complex - see JSON report)

ifn_gamma_count: 0

total_predicted: 6

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 9

successful_models: 9

pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 15

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 8

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:

[
"ETNYKAVSYHASGHS",
"EETNYKAVSYHASGH"
]

mhc1_epitopes_used: [

Multi-Epitope Vaccine (MEV) Design Report

"RYTFGAALF",
"ETNYKAVSY",
"HASGHSVAY",
"NPVTSVFQY"
]

bcell_epitopes_used: []

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAETNYKAVSYHASGHSGPGPGEETNYKAVSYHASGHRYTFGAALFAAYETNYKAVSYAAYHASGHSVAYAAYNPVTSVFQYEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 129

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (387 chars - see JSON report)

dna_length_bp: 387

cai_score: 0.9029

gc_content_pct: 59.69

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 387

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:

{
"vector": "pShuttle-CMV",
"insert_size_bp": 387,
"cloning_sites": {
"upstream": "NotI (5')",
"downstream": "XhoI (3')"
},
"promoter": "CMV immediate-early",
"terminator": "BGH poly-A signal",
"orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}

Overall MEV Candidacy Assessment

STRONG MEV CANDIDATE